future for particle physics. The standard model was in place and its remarkable success at predicting experimental outcomes indicated that its verification was merely a matter of time and details. Going beyond its limits to include gravity and possibly to explain the experimental input on which it relies the 19 numbers summarizing the elementary particle masses, their force charges, and the relative strengths of the forces, numbers that are known from experiment but are not understood theoretically—was so daunting a task that all but the most courageous physicists recoiled at the challenge. But six months later the mood had swung completely around. The success of Green and Schwarz finally trickled down even to first-year graduate students, and an electrifying sense of being on the inside of a profound moment in the history of physics displaced the previous ennui. A number of us consistently worked deep into the night to try to master the vast areas of theoretical physics and abstract mathematics that are required to understand string theory.

The period from 1984 to 1986 has come to be known as the “first superstring revolution.” During those three years more than a thousand research papers on string theory were written by physicists from around the world. These works showed conclusively that numerous features of the standard model—features that had been painstakingly discovered over the course of decades of research—emerged naturally and simply from the grand structure of string theory. As Michael Green has said, “The moment you encounter string theory and realize that almost all of the major developments in physics over the last hundred years emerge—and emerge with such elegance—from such a simple starting point, you realize that this incredibly compelling theory is in a class of its own.”5 Moreover, for many of these features, as we shall discuss, string theory offers a far fuller and more satisfying explanation than is found in the standard model. These developments convinced many physicists that string theory was well on its way to fulfilling its promise of being the ultimate unified theory.

Nonetheless, over and over again string theorists encountered a significant stumbling block. In theoretical physics research, one is frequently confronted with equations that are just too hard to understand or to analyze. Typically, physicists don’t give up, but try to solve the equations approximately. The situation in string theory is